

EstimateX

Literature Review — Reading Compilation

Predicting Product Development Time and Cost using Machine Learning

This document summarizes existing published research relevant to our project, to be read by the full team before we begin drafting our own research paper. Deep learning based studies have been excluded, since this project uses classical Machine Learning only. A note at the end explains that decision.

Important: the summaries below are written in our own words from each paper's abstract/summary — they are not copied text from the papers (reproducing full copyrighted text is not something this document does). Where a paper's full text could not be reliably accessed, only the citation and a short note is given — read that paper directly via its link.

Section 1 — Core ML Papers on Software Effort / Cost Estimation

1. Predicting Effort, Duration and Cost in Software Development Using Machine Learning Technique

JARDCS, 2019

What it did:

Applied regression-based machine learning models to predict software project effort, duration, and cost from historical project data.

Key finding:

Reported that ML-based prediction gave more accurate results than traditional formula-based models such as COCOMO.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://www.jardcs.org/abstract.php?id=665>

2. Software Effort Estimation Using Machine Learning Methods

Oliveira et al., presented at ISCIS 2007, published via ResearchGate

What it did:

Built machine learning models — back-propagation neural networks, regression trees, radial basis function (RBF) networks, and support vector regression — combined with Principal Component Analysis (PCA) for dimensionality reduction, to estimate software effort.

Datasets used:

Three project datasets: NASA (60 projects), USC (63 projects), and a Software Data Repository of Turkish software organizations (23 projects).

Key findings:

Regression trees performed best on the NASA and Turkish datasets (PRED(30) above 75%), while RBF networks with PCA performed best on the USC dataset. All ML methods clearly outperformed the COCOMO parametric model, especially on the Turkish dataset. The authors concluded that no single algorithm works best across every dataset — the right method depends on the specific project data.

Link: https://www.researchgate.net/publication/4321088_Software_effort_estimation_using_machine_learning_methods

3. Analysis of Software Effort Estimation by Machine Learning Techniques

IETA, International Information and Engineering Technology Association, 2023

What it did:

Compared multiple classical machine learning algorithms — including Random Forest, Gradient Boosting, and Support Vector Regression — against each other on the same software effort estimation task.

Key finding:

Found that ensemble methods, particularly Random Forest, consistently gave the strongest and most reliable accuracy among the models tested.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://www.iieta.org/download/file/fid/114839>

4. Enhancing Software Cost Estimation Using Feature Selection and Machine Learning Techniques

Computers, Materials & Continua (CMC), 2024

What it did:

Studied how different feature selection techniques affect the accuracy of machine learning models used for software cost estimation.

Dataset used:

The COCOMO-NASA dataset, containing 93 software projects described by 24 project attributes.

Methodology:

Applied several machine learning algorithms in combination with three different feature selection methods to remove redundant/less useful input features before training.

Key finding:

Principal Component Analysis (PCA) was the most effective feature selection method of the three tested, giving the highest precision, accuracy, and recall — showing that careful feature selection can meaningfully improve prediction quality, sometimes more than the choice of algorithm itself.

Link: <https://www.sciopen.com/article/10.32604/cmc.2024.057979>

5. A Survey of Machine Learning Approach to Software Cost Estimation

IEEE, 2021

What it covers:

A survey-style paper reviewing multiple machine learning techniques that have been applied to software cost estimation over time, comparing their reported strengths and weaknesses.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://ieeexplore.ieee.org/document/9491912/>

6. Machine Learning Models for Software Cost Estimation

Al-Asheeri & Hammad, IEEE, 2019

What it covers:

Evaluates and compares different machine learning models applied specifically to the task of software cost estimation, aiming to identify which models are most suitable for this problem.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://ieeexplore.ieee.org/document/8910327/>

Section 2 — Broader Project Cost Estimation (Multi-Domain Evidence)

These papers are not about software specifically — they cover project cost estimation with AI/ML across construction and other industries. They are useful evidence for our project's multi-domain angle, showing that the same core idea (learning from past project data to predict cost) is applied well beyond software.

7. Advancement of Artificial Intelligence in Cost Estimation for Project Management Success: A Systematic Review of Machine Learning, Deep Learning, Regression, and Hybrid Models

MDPI, 2024 — Systematic review of 39 peer-reviewed articles (2016–2024), selected via PRISMA from an initial pool of 917 articles

Scope:

Covers cost estimation across multiple sectors — construction, healthcare, manufacturing, and real estate — not just software.

Model categories reviewed:

Deep Learning (CNN, RNN, LSTM, DNN), Machine Learning (ANNs, SVMs, decision trees, gradient boosting), Regression models, and Hybrid models.

Key findings:

Machine learning models reached 75-80% accuracy and regression models 70-80%, while deep learning/hybrid models reached 85-90% — but at the cost of needing far more data and being harder to interpret. ANNs were the most versatile ML technique across industries, and XGBoost performed especially well in construction specifically.

Link: <https://www.mdpi.com/2673-3951/6/2/35>

8. Applications of Machine Learning and Artificial Intelligence in Construction Project Cost Prediction: A Scientometric Analysis and Qualitative Review

Frontiers in Built Environment, 2026 — 138 core journal articles analyzed, 1994-2025

What it found:

Research in this area has grown rapidly since 2021 (over 76% of all publications on this topic happened between 2021-2025). Methods have shifted from single models like ANN/SVM toward ensemble methods (Random Forest, XGBoost) and deep learning.

Key limitation noted:

Despite better accuracy, practical use is still limited by unstable data quality, poor generalization across regions, and low interpretability of the models — the same kind of "explainability" gap our own project is trying to address.

Link: <https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2026.1867673/full>

9. Cost Estimation and Prediction in Construction Projects: A Systematic Review on Machine Learning Techniques

Discover Applied Sciences (Springer), 2020 — Systematic review of 92 papers, 1985-2020

What it found:

Artificial Neural Networks were the dominant machine learning method used for construction cost estimation, with Regression Analysis as the second most common. Hybrid models (ANN combined with fuzzy logic, genetic algorithms, or case-based reasoning) generally outperformed plain ANN models.

Key finding relevant to us:

The number of input features mattered a lot — adding more input factors than necessary actually reduced model accuracy, reinforcing the value of careful feature selection over throwing in every possible variable.

Link: <https://link.springer.com/article/10.1007/s42452-020-03497-1>

10. Machine Learning Algorithms for Constructions Cost Prediction: A Systematic Review

International Journal of Nonlinear Analysis and Applications

What it covers:

A systematic review specifically of machine learning algorithms (as opposed to older statistical/expert-judgement methods) used for predicting construction project costs.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: https://ijnaa.semnan.ac.ir/article_6693_45f00920004af5c5ee3b55c9a876199b.pdf

11. Transparent and Reliable Construction Cost Prediction Using Advanced Machine Learning and Explainable AI

ScienceDirect, 2025

What it covers:

Combines machine learning cost prediction with Explainable AI (XAI) methods, aiming to make construction cost predictions both accurate and interpretable — directly relevant to our own "explainability" objective.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://www.sciencedirect.com/science/article/pii/S2215098625002149>

12. Using Machine Learning to Predict Cost Overruns in Construction Projects

ResearchGate, 2023

What it covers:

Focuses specifically on predicting cost overruns (projects going over budget) in construction using machine learning models trained on historical project data.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: https://www.researchgate.net/publication/371487015_Using_Machine_Learning_to_Predict_Cost_Overruns_in_Construction_Projects

13. Predicting Construction Costs with Machine Learning: A Comparative Study on Ensemble and Linear Models

Journal of Electronics, Electromedical Engineering, and Medical Informatics

What it covers:

Compares ensemble machine learning models (like Random Forest/Gradient Boosting) against simple linear models for predicting construction costs.

Note: full text is paywalled/inaccessible — the description above is based on the paper's title and published abstract snippet only. Read the full paper via the link before citing it in detail.

Link: <https://jeeemi.org/index.php/jeeemi/article/view/799>

Section 3 — Why We Chose Machine Learning and Not Deep Learning

Across nearly every paper reviewed above — including the ones directly about software cost estimation and the broader multi-domain ones — a consistent pattern shows up: deep learning models (CNNs, RNNs, LSTMs, DNNs) report slightly higher accuracy on paper (roughly 85-90% in the MDPI review, for example, versus 75-80% for classical ML), but that advantage comes with real costs that don't fit our project.

Small datasets:

Every core dataset used in this exact research area — COCOMO-NASA (93 projects), Desharnais (81 projects), Maxwell, ISBSG — has only a few dozen to a few hundred records. Deep learning models need far larger datasets to learn reliably; on data this small, they tend to overfit rather than genuinely learn patterns, while classical ML models like Random Forest and Gradient Boosting are specifically well-suited to small, structured, tabular data like this.

Explainability matters more than a few extra accuracy points:

One of our core objectives is showing **why** a prediction was made — which project factors drove the estimate. Classical ML models (especially tree-based ones) make this straightforward through feature importance. Deep learning models are much harder to interpret, which directly works against our explainability goal — a pattern the Frontiers (2026) and ScienceDirect XAI papers above call out as an open problem in this field, not something deep learning has solved.

Practicality for a multi-domain, real-world tool:

Deep learning requires more computing power, more tuning, and more data pipeline engineering to deploy reliably. Since our goal is a usable, accessible tool — not a research benchmark — classical ML keeps the system lightweight, easier to train on modest data from any domain, and easier to retrain as new data comes in.

The accuracy gap is modest, not dramatic:

The MDPI review shows only a 5-10 percentage point accuracy gap between ML and DL/hybrid models in this exact application area. Given the added complexity, data requirements, and loss of interpretability, that modest gain isn't worth the tradeoff for a project of this scale.

In short: deep learning is not being avoided because it doesn't work — the literature shows it can work, given enough data. It's being set aside because our available data is small, our core objective is explainability, and our goal is a practical, deployable tool rather than squeezing out the last few points of benchmark accuracy.